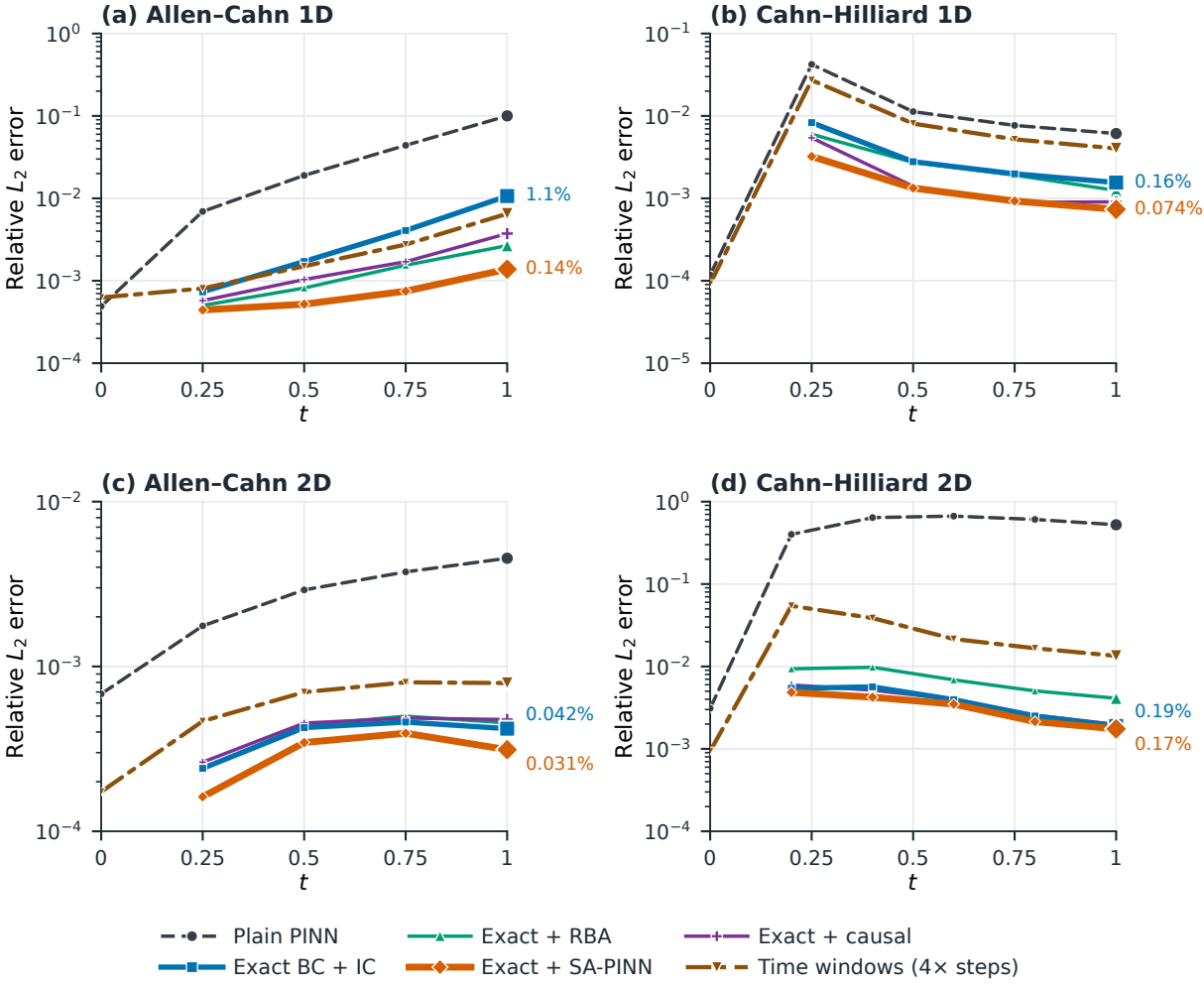
(a) Allen–Cahn 1D
(b) Cahn–Hilliard 1D
(c) Allen–Cahn 2D
(d) Cahn–Hilliard 2D
Relative L_2 error
t
1.1%
0.14%
0.16%
0.074%
0.042%
0.031%
0.19%
0.17%
Plain PINN
Exact BC + IC
Exact + RBA
Exact + SA-PINN
Exact + causal
Time windows (4× steps)